DAYCLI Guidelines

Guidelines for the International Exchange of Daily Climate Data

Supporting document for SERCOM Management Group approval

1. Purpose

The *DAYCLI Guidelines — Guidelines for the International Exchange of Daily Climate Data* provide conceptual and practical guidance to National Meteorological and Hydrological Services (NMHSs) on the production, interpretation and international exchange of official daily climate data through the DAYCLI message.

The Guidelines address the climatological requirements and information needed to preserve the meaning and integrity of national daily climate datasets, while supporting consistent interpretation and interoperability at the international level.

2. Background and rationale

Daily climate observations are essential for climate monitoring, the analysis of climate variability and change, the identification and assessment of climate extremes, and the development of climate services. However, official daily climate values are produced according to national climatological practices that may differ in their definitions of the climatological day, attribution day, calculation methods, quality control procedures and other operational aspects.

Consequently, international exchange requires more than the transmission of numerical values. Information on temporal references, observing stations, data processing and quality is needed to ensure that exchanged data can be correctly interpreted, compared and reused beyond their national context.

The DAYCLI message was developed to support the routine international exchange of officially validated national daily climate data together with the information required for their correct interpretation. The Guidelines provide a common conceptual and practical framework for this purpose, helping Members improve interoperability while preserving the meaning and integrity of their national climate records.

3. Development, institutional contributions and technical status

The Guidelines have been developed under the leadership of the WMO Services Commission (SERCOM), through its Standing Committee on Climate Services (SC-CLI), with substantive contributions from the Expert Team on Data Development and Stewardship (ET-DDS).

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The development of DAYCLI has also involved close collaboration with the WMO Infrastructure Commission (INFCOM), particularly its Standing Committee on Information Management and Technology (SC-IMT) and the Expert Team on Data Standards (ET-Data Standards). Their contribution has been essential to the development, validation and maintenance of the DAYCLI BUFR sequence, based on the DAYCLI information model.

The DAYCLI BUFR sequence **3 07 095**, corresponding to the version described in these Guidelines, was approved by INFCOM through the fast-track procedure for amendments to the *Manual on Codes* (WMO-No. 306), Volume I.2, under **FT2026-2 on 4 March 2026**.

The Guidelines complement the technical specifications of the DAYCLI message in WMO-No. 306. They explain the climatological concepts, information requirements and implementation considerations needed for the consistent interpretation and international exchange of daily climate data. They are intended as practical, non-regulatory guidance and do not seek to impose uniform national climatological practices.

4. Consultation and review

On 26 May 2026, Qingchen Chao, Chair of the SERCOM Standing Committee on Climate Services (SC-CLI), officially circulated the draft DAYCLI Guidelines for review on behalf of SERCOM/SC-CLI.

The draft was shared with the relevant Standing Committees and Expert Teams of SERCOM and INFCOM, as well as other international and regional organizations. Reviewers were invited to comment on readability, practical implementation, consistency with BUFR specifications and the treatment of metadata, with feedback requested by 19 June 2026.

The *Acknowledgements* section of the Guidelines recognizes the contributions of reviewers from Météo-France, Météo-France International, CIMA (Argentina), the German Weather Service (DWD) and the Australian Bureau of Meteorology. Their comments and recommendations helped improve the technical quality, clarity and consistency of the document. The Guidelines also acknowledge the technical contributions of the WMO Secretariat and experts involved in the development and validation of the DAYCLI BUFR representation.

This process brought together climatological expertise, operational implementation experience and technical knowledge of international data representation.

5. Expected benefits for WMO Members

The Guidelines are expected to:

- Support NMHSs in implementing and operationally using DAYCLI for the exchange of official daily climate data;
- Improve the consistent interpretation and international reuse of daily climate observations;

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- Preserve the climatological meaning and integrity of national daily climate datasets by documenting the temporal references, station characteristics, processing information and quality information needed for interpretation;
- Facilitate interoperability between national climate data management systems and international data exchange arrangements;
- Support climate monitoring, research, the analysis of climate extremes and the development of climate services.

The Guidelines follow a progressive approach to implementation, allowing Members to establish the essential information requirements first and enrich the information exchanged as national capabilities and available metadata permit.

6. Proposed action

The SERCOM Management Group is invited to:

1. **Approve the DAYCLI Guidelines — Guidelines for the International Exchange of Daily Climate Data** as practical, non-regulatory guidance supporting the production, interpretation and international exchange of official daily climate data by WMO Members;
2. Recognize their contribution to preserving the climatological meaning and integrity of national daily climate datasets while facilitating interoperability and international reuse;
3. Support the necessary subsequent steps towards final document verification, publication and dissemination by WMO in accordance with the applicable procedures.

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